SQL Project

Airline Ticket Price Analysis

Objective

The objective of this project is to analyze airline ticket pricing patterns based on factors such as airline names, travel class, route distance, and booking timing. The analysis aims to identify trends that influence ticket prices and help understand airline pricing strategies.

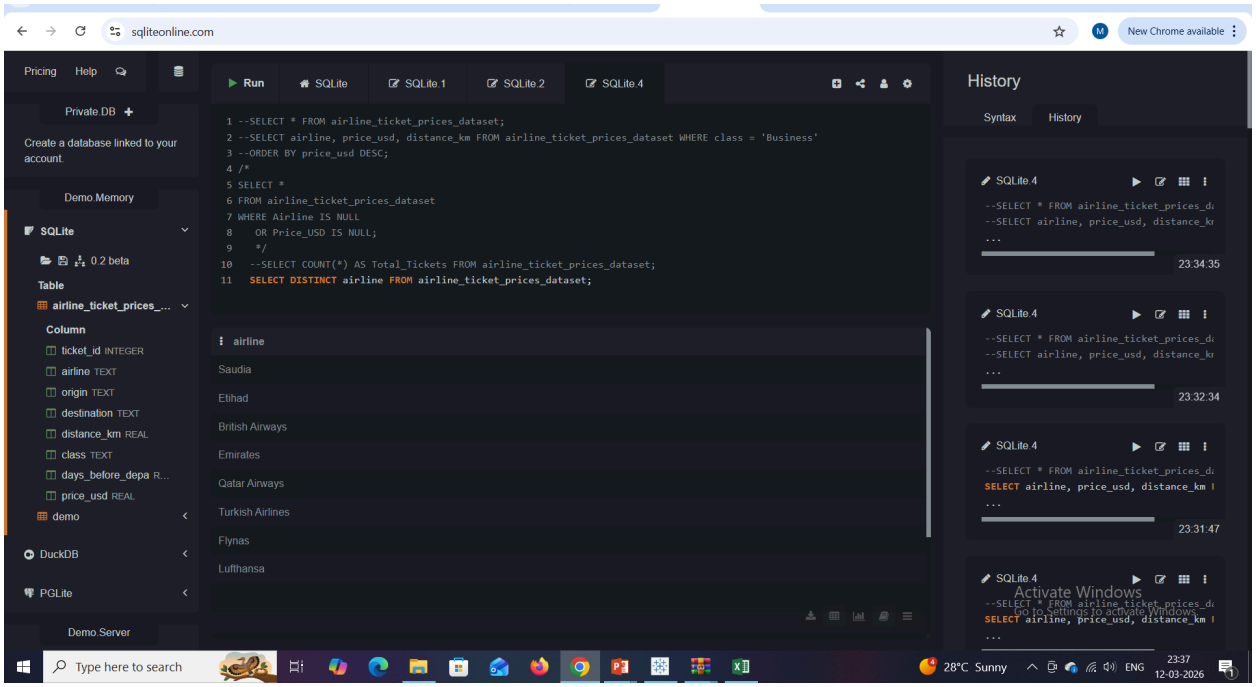
Data Preparation

The dataset contains airline ticket booking information including airline name, origin and destination cities, travel distance, ticket class, number of days before departure when the ticket was booked, and ticket price in USD.

The dataset was imported into SQLite where SQL queries were used to explore and analyze the data. Numeric fields such as ticket price and distance were converted into appropriate data types to enable calculation.

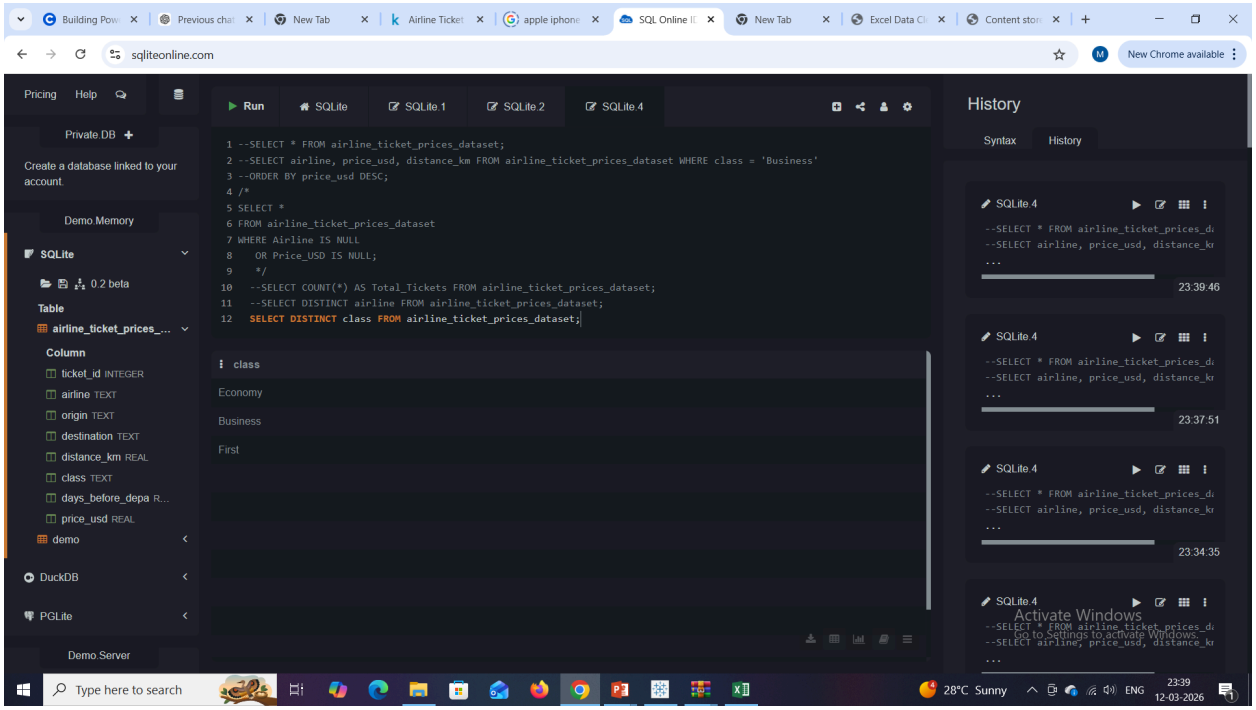
Q1: List the distinct Airlines from the dataset.

Query: SELECT DISTINCT airline FROM airline_ticket_prices_dataset;



Q2: List the distinct Travel class from the dataset.

Query: SELECT DISTINCT class FROM airline_ticket_prices_dataset;



Q3: Average ticket price by airline

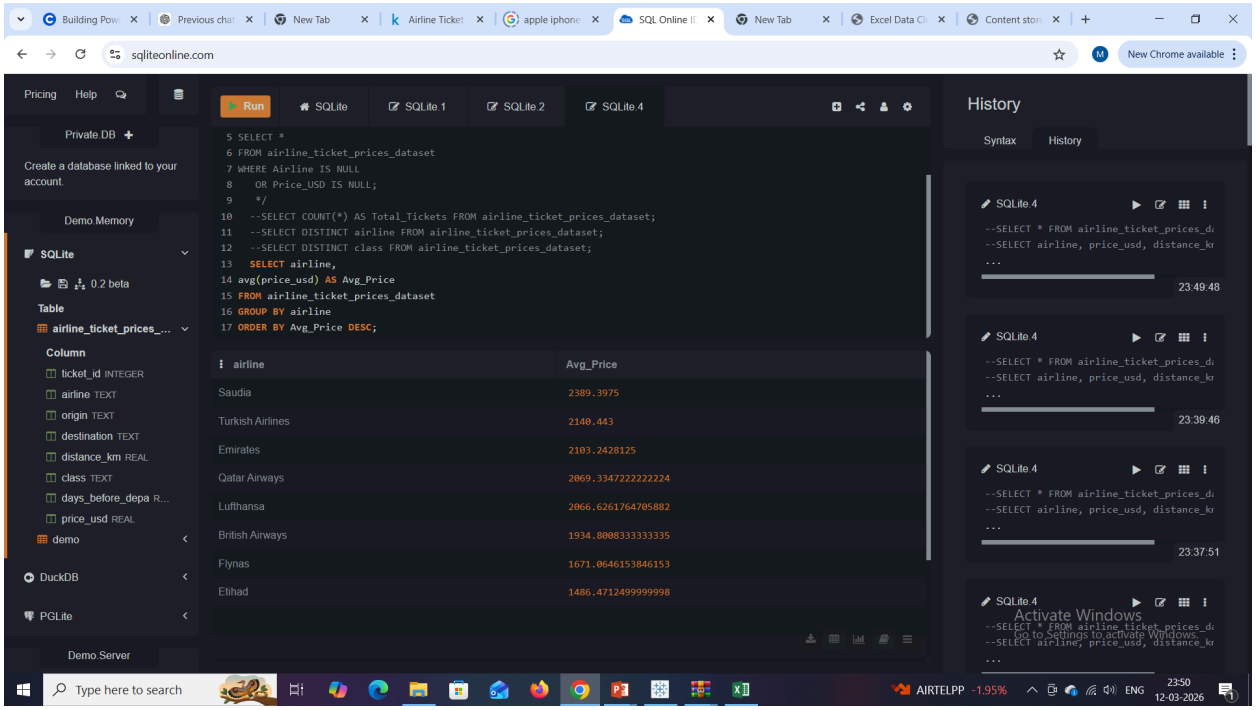
Query: SELECT airline,

avg(price_usd) AS Avg_Price

FROM airline_ticket_prices_dataset

GROUP BY airline

ORDER BY Avg_Price DESC;



Q4: Average ticket price by class

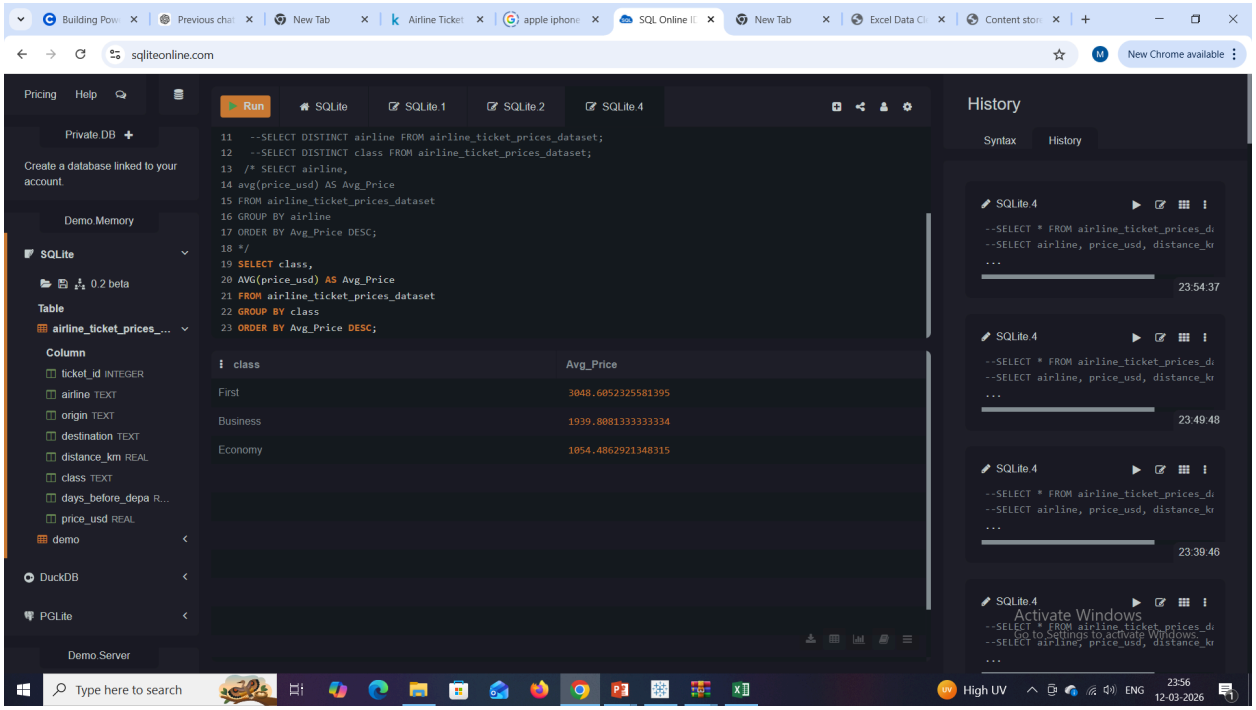
Query: SELECT class,

AVG(price_usd) AS Avg_Price

FROM airline_ticket_prices_dataset

GROUP BY class

ORDER BY Avg_Price DESC;



Q5: Most expensive Routes

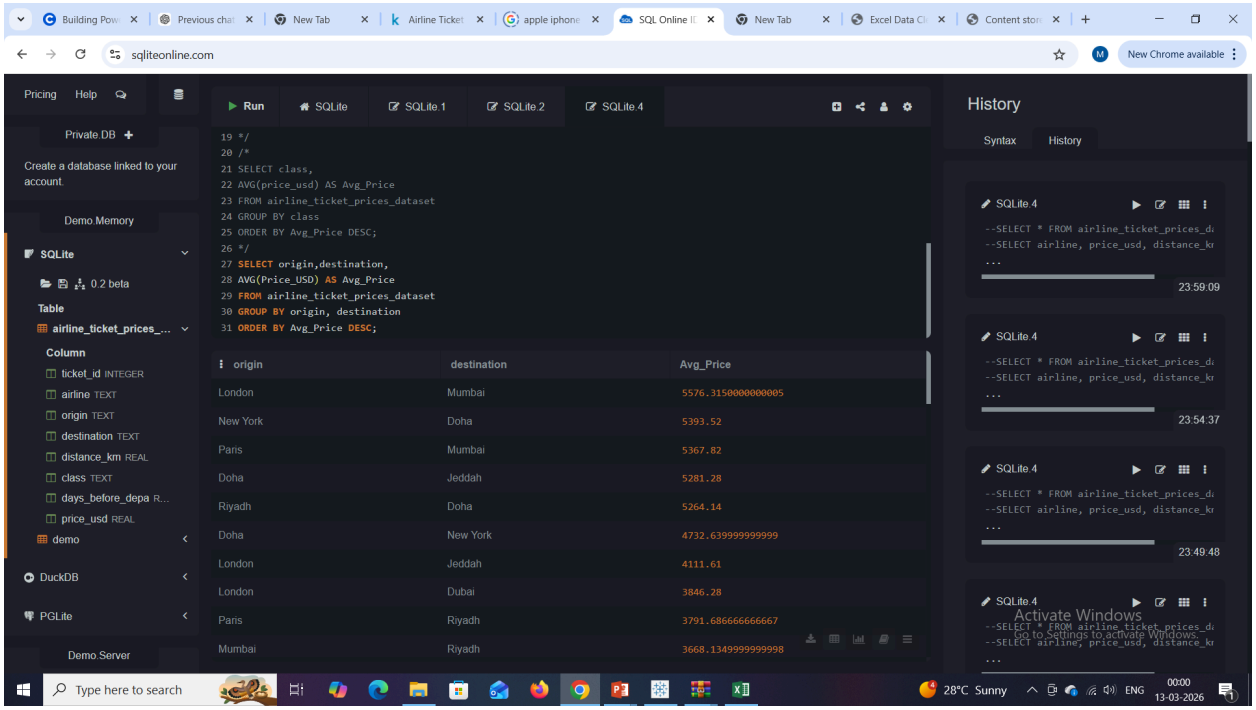
Query: SELECT origin,destination,

AVG(Price_USD) AS Avg_Price

FROM airline_ticket_prices_dataset

GROUP BY origin, destination

ORDER BY Avg_Price DESC;



Distance vs Price Analysis

Q6:Check if longer flights cost more.

SELECT airline,

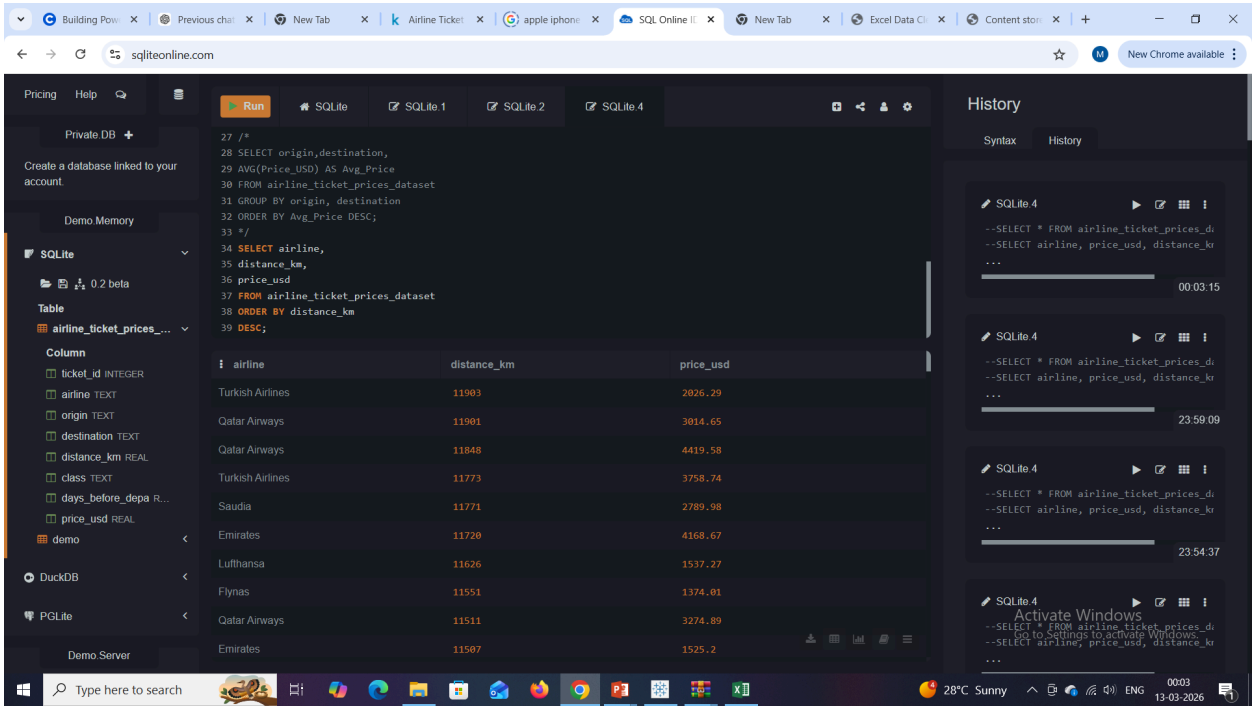
distance_km,

price_usd

FROM airline_ticket_prices_dataset

ORDER BY distance_km

DESC;



Booking Behavior Analysis

Q7:Do early bookings cost more?

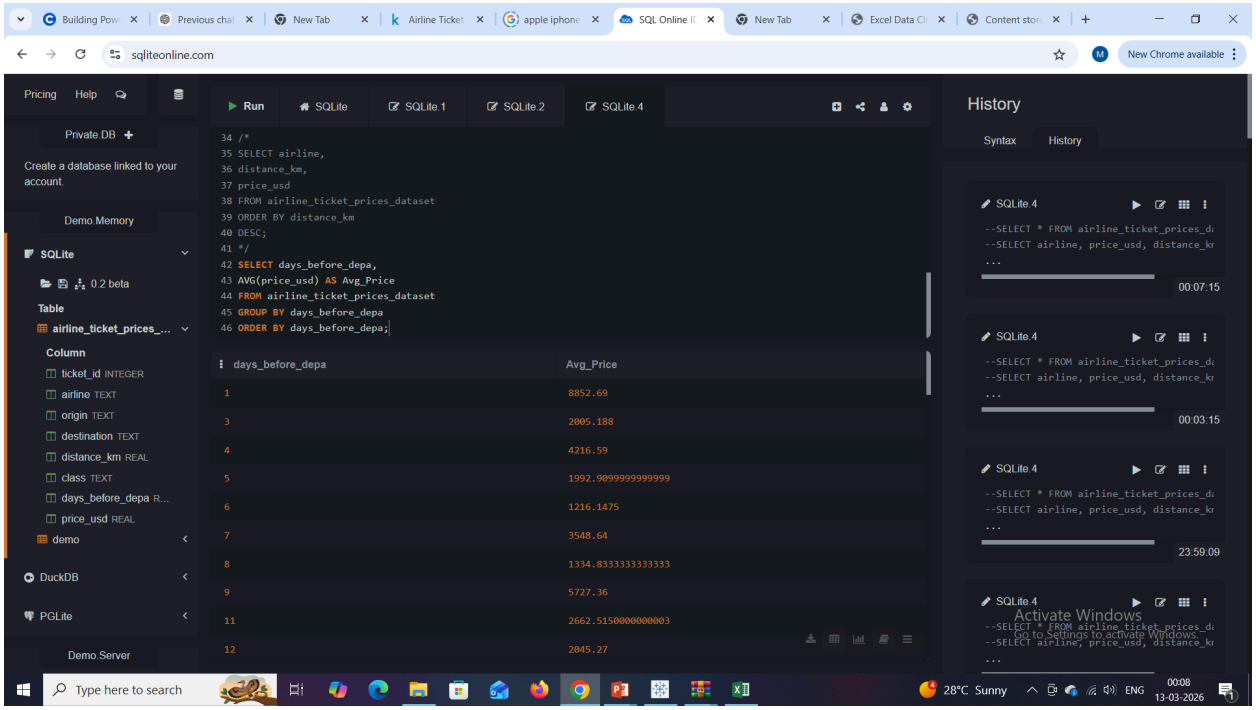
Query: SELECT days_before_depa,

AVG(price_usd) AS Avg_Price

FROM airline_ticket_prices_dataset

GROUP BY days_before_depa

ORDER BY days_before_depa;



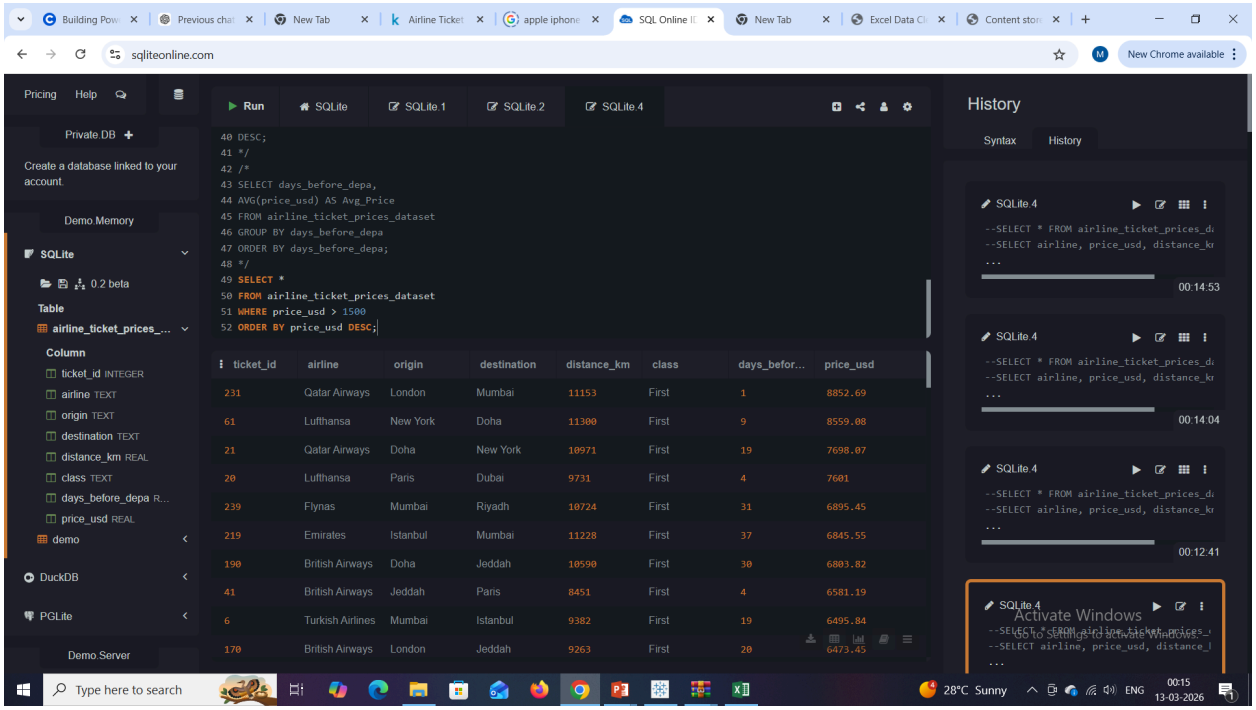
Premium flights

Q8: Find expensive flights.

Query: FROM airline_ticket_prices_dataset

WHERE price_usd > 1500

ORDER BY price_usd DESC;

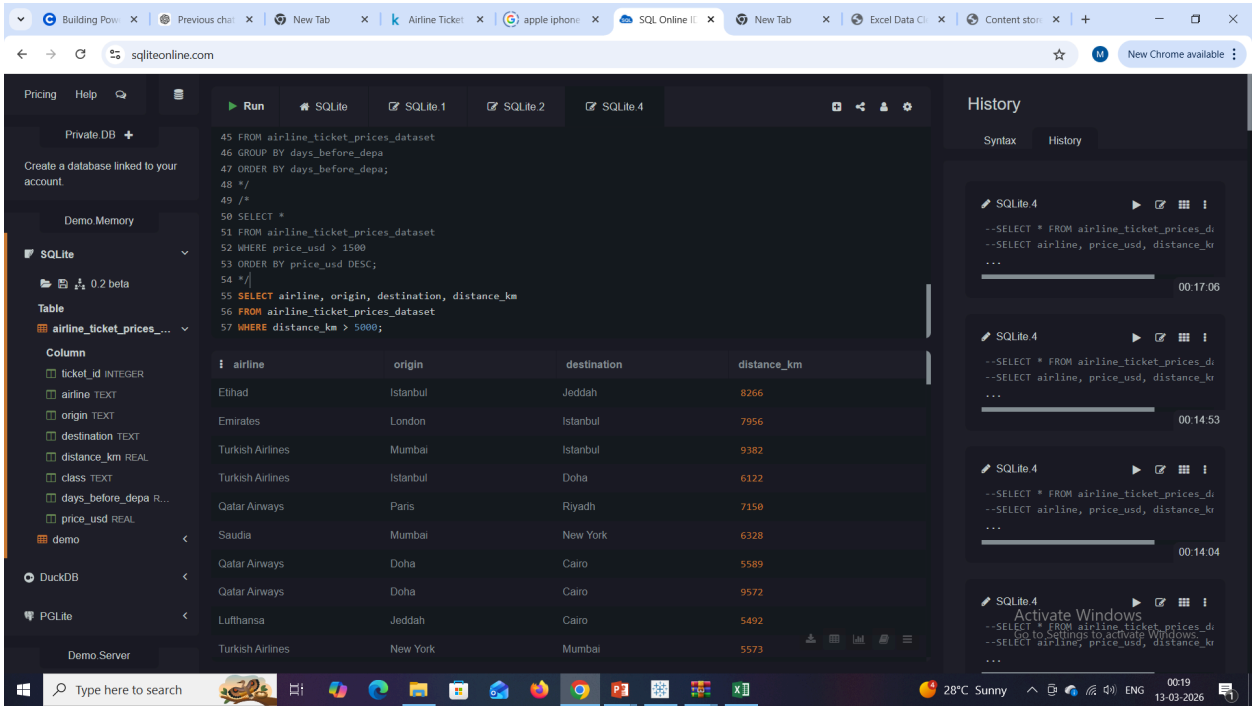


Q9: Find the long distance flights.

Query: SELECT airline, origin, destination, distance_km

FROM airline_ticket_prices_dataset

WHERE distance_km > 5000;



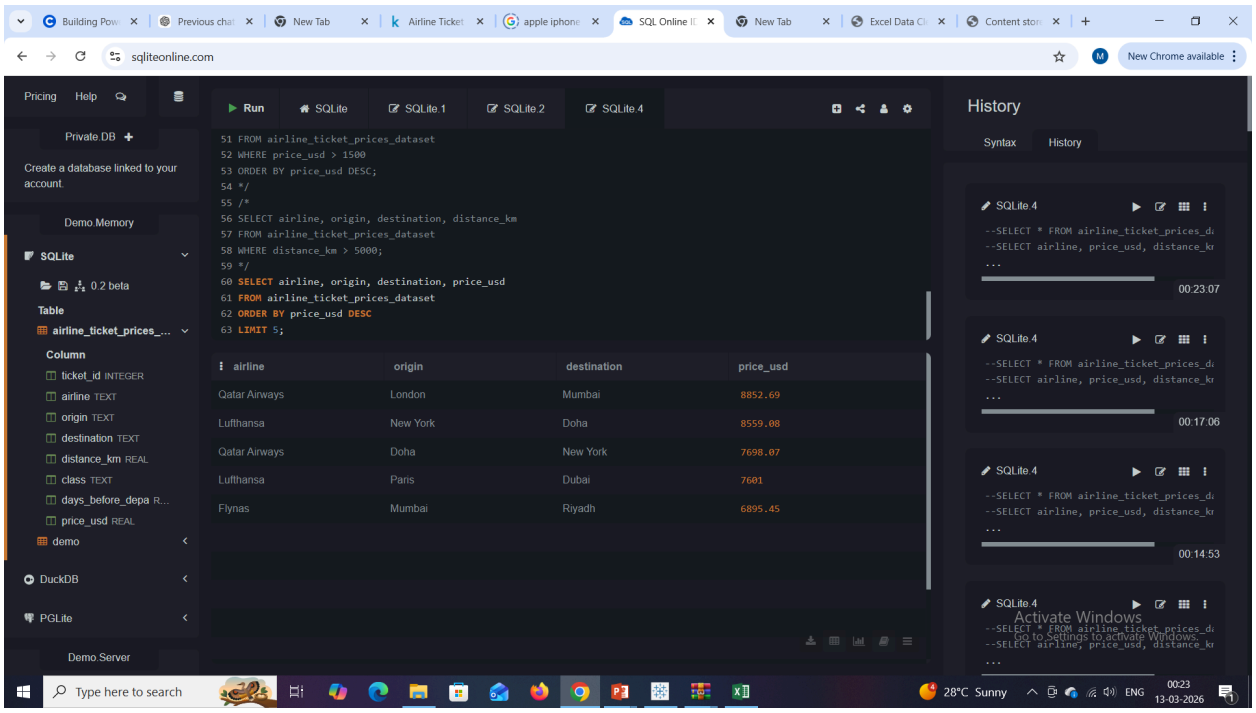
Q10: Find the top **5** expensive flights.

Query: SELECT airline, origin, destination, price_usd

FROM airline_ticket_prices_dataset

ORDER BY price_usd DESC

LIMIT 5;



Comparison of Price in different travel class among different Airlines

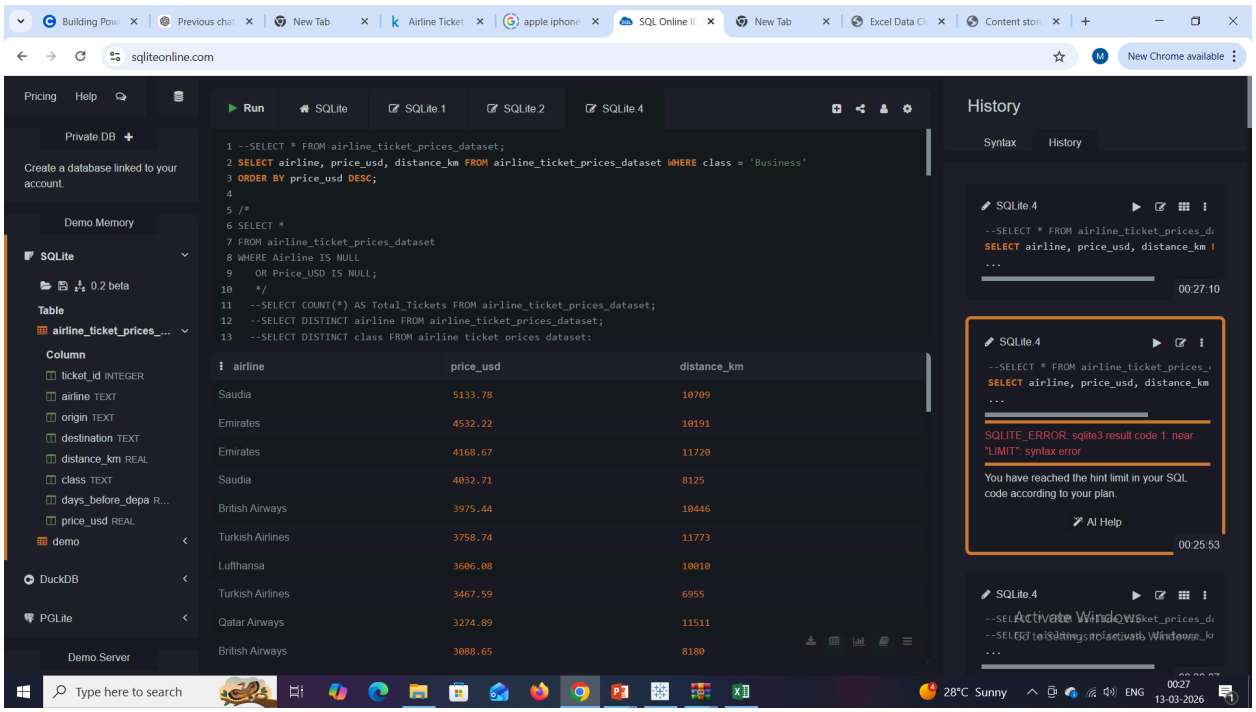
Q11: Business class price analysis.

Query: SELECT airline, price_usd, distance_km

FROM airline_ticket_prices_dataset

WHERE class = 'Business'

ORDER BY price_usd DESC;



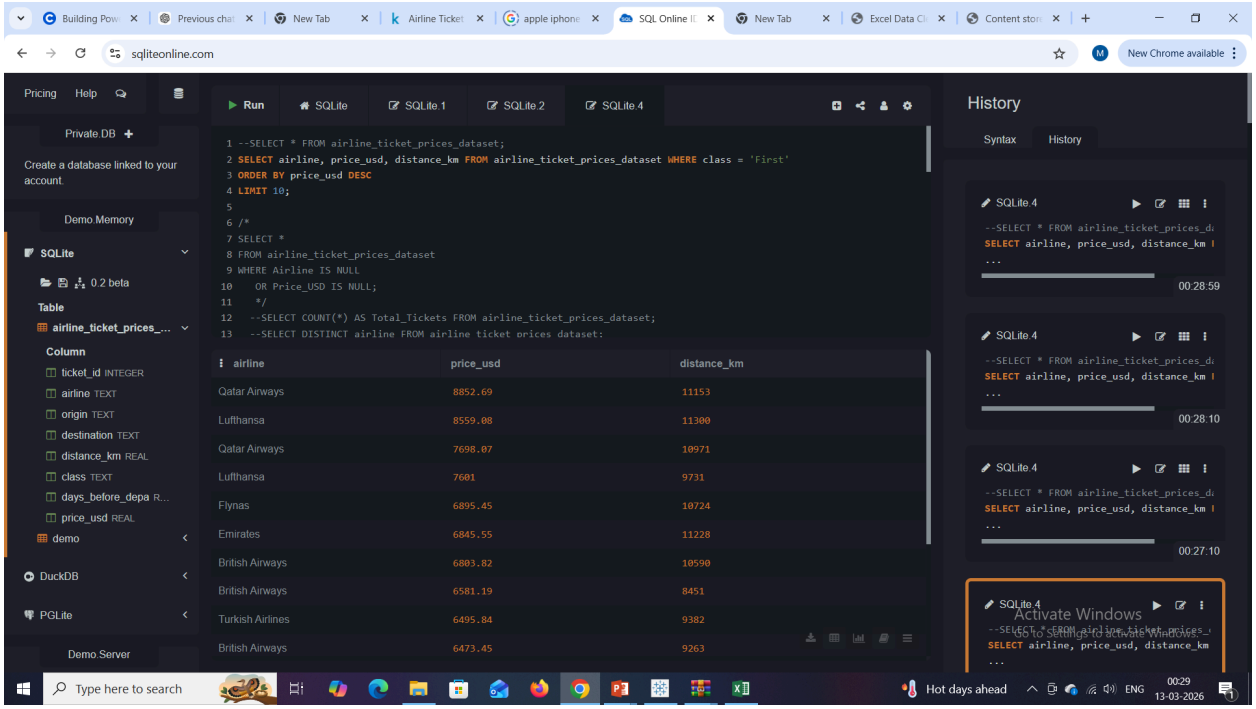
Q12: ‘First class’ price analysis in different Airlines

Query: SELECT airline, price_usd, distance_km

FROM airline_ticket_prices_dataset

WHERE class = 'First'

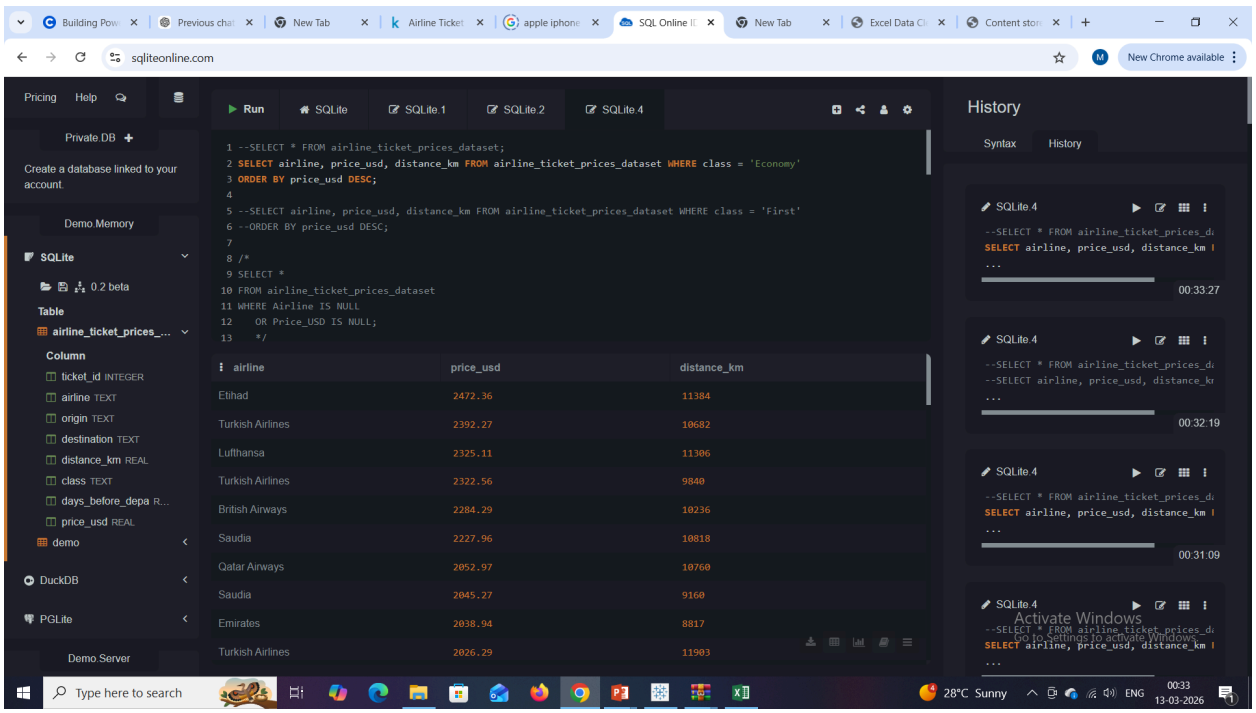
ORDER BY price_usd DESC;



Q13: Economy class price analysis in different Airlines.

Query: SELECT airline, price_usd, distance_km

FROM airline_ticket_prices_dataset

WHERE class = 'Economy'

ORDER BY price_usd DESC;